

CompTIA

A+ Core 1

Exam 220-1201

A+ 220-1201

Module 2



Installing Motherboards and Connectors

Prepare for A+ Core 1 by:

- Explaining cable types and connectors
- Installing and configuring motherboards
- Explaining legacy cable types

Lesson 2.1

Cables and Connectors

Personal Computers (Slide 1 of 3)

Components
housed within the
system case

Peripherals- external
components that
add functionality to
system

Personal Computers (Slide 2 of 3)

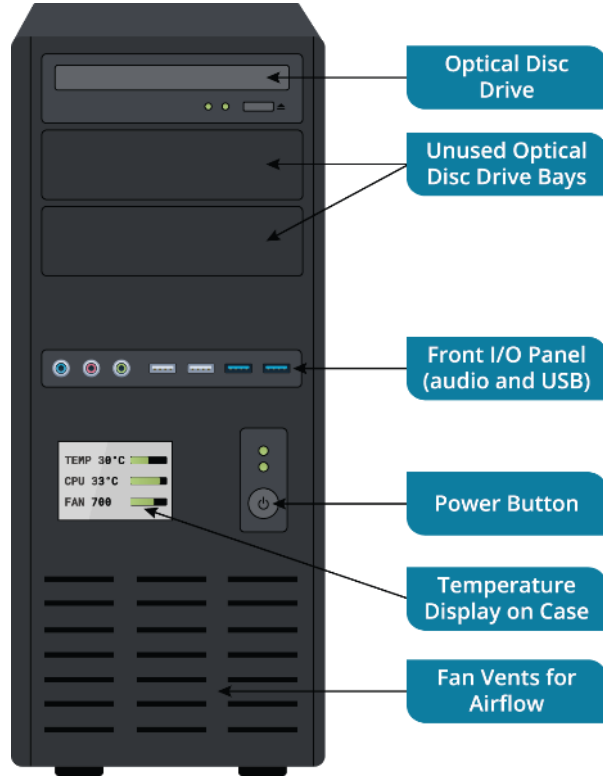


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- Front
 - Access to disk drives, power switch, and LEDs
 - May also have peripheral ports
 - Air vents- Keep clean

Personal Computers (Slide 3 of 3)

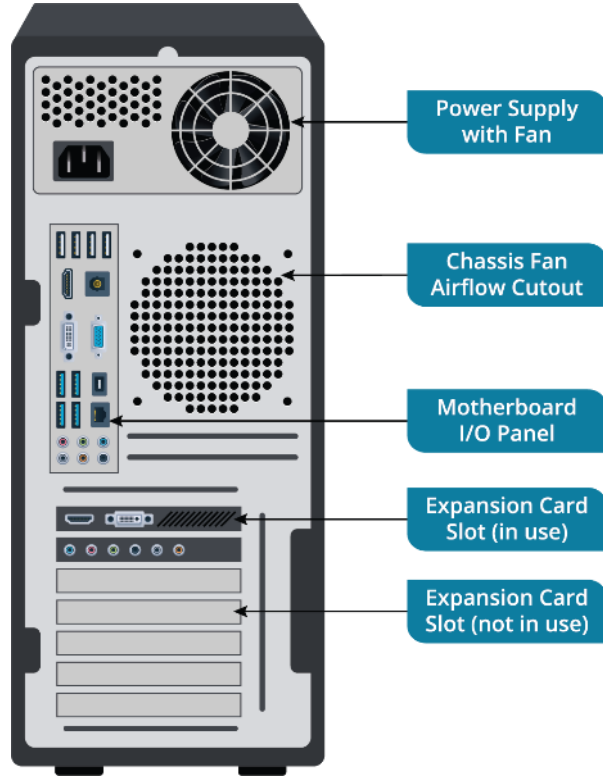


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- Rear
 - Access to power plug, power supply, rear peripheral ports
 - May also have expansion card access
 - Air Vents- exhaust hot air and dust from system; Keep clean

Peripheral Devices (Slide 1 of 2)



Interfaces, Ports, Connectors



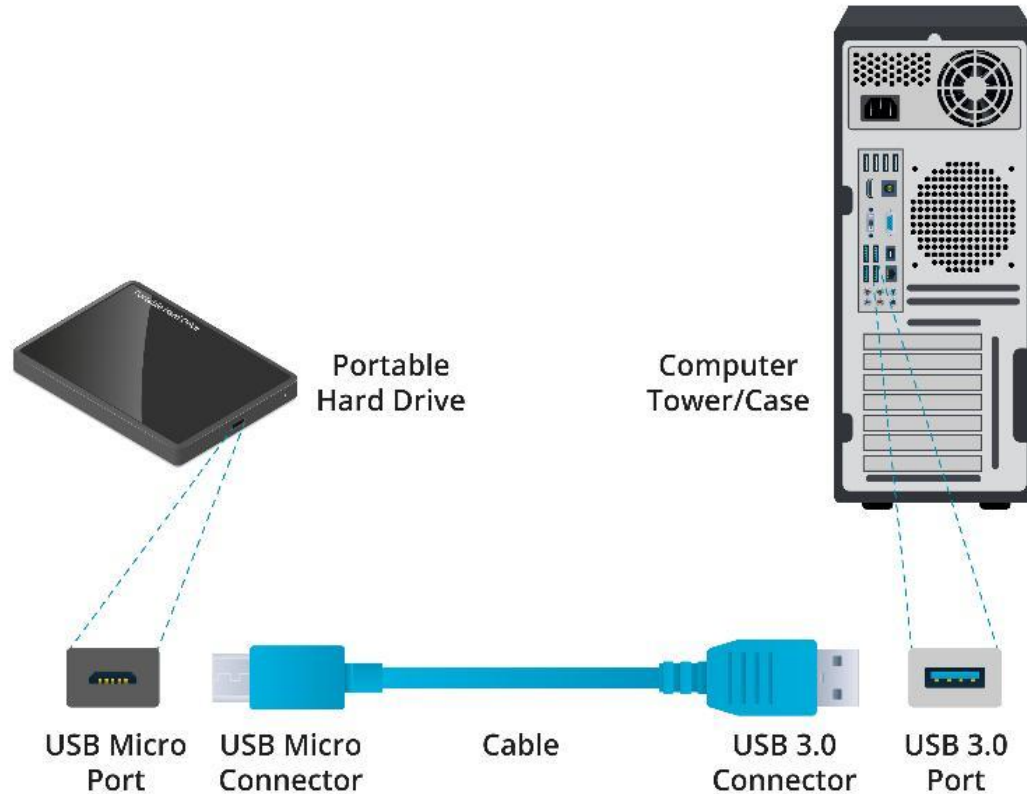
Data Storage and Transfer Unit of Measure

Bytes vs Bits

Conversions between Units of Measure

- Kilo to Mega to Giga to Tera to Peta

Peripheral Devices (Slide 2 of 2)



Universal Serial Bus Cables (Slide 1 of 3)



- Standard for connecting modern peripherals
- Audio, Video, Printers, Storage, Input and Output
- Power and Data
- Length- 3m/5m

Universal Serial Bus Cables

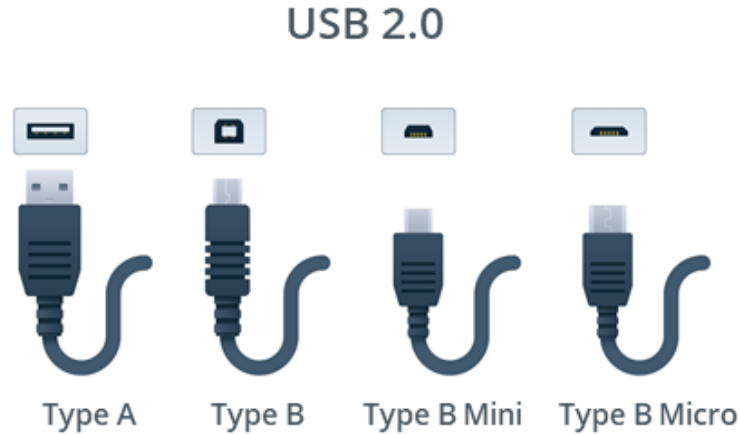


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- Cable connectors can be same on each end or provide conversion between connector types
- Slower data speed than 3.0

Universal Serial Bus Cables

- Faster data transfer and higher power
- USB C is being mandated for mobile electronics including smartphones and tablets



Image © 123RF.com

USB Standards

1.0/1.1- 1.5 Mbps or 12 Mbps

2.0- 480 Mbps

3.0- 5 Gbps

3.1- 10 Gbps

3.2- Gen 1/2/2x2- 5 Gbps, 10 Gbps, 20 Gbps

4.0- 40 Gbps

Display Types

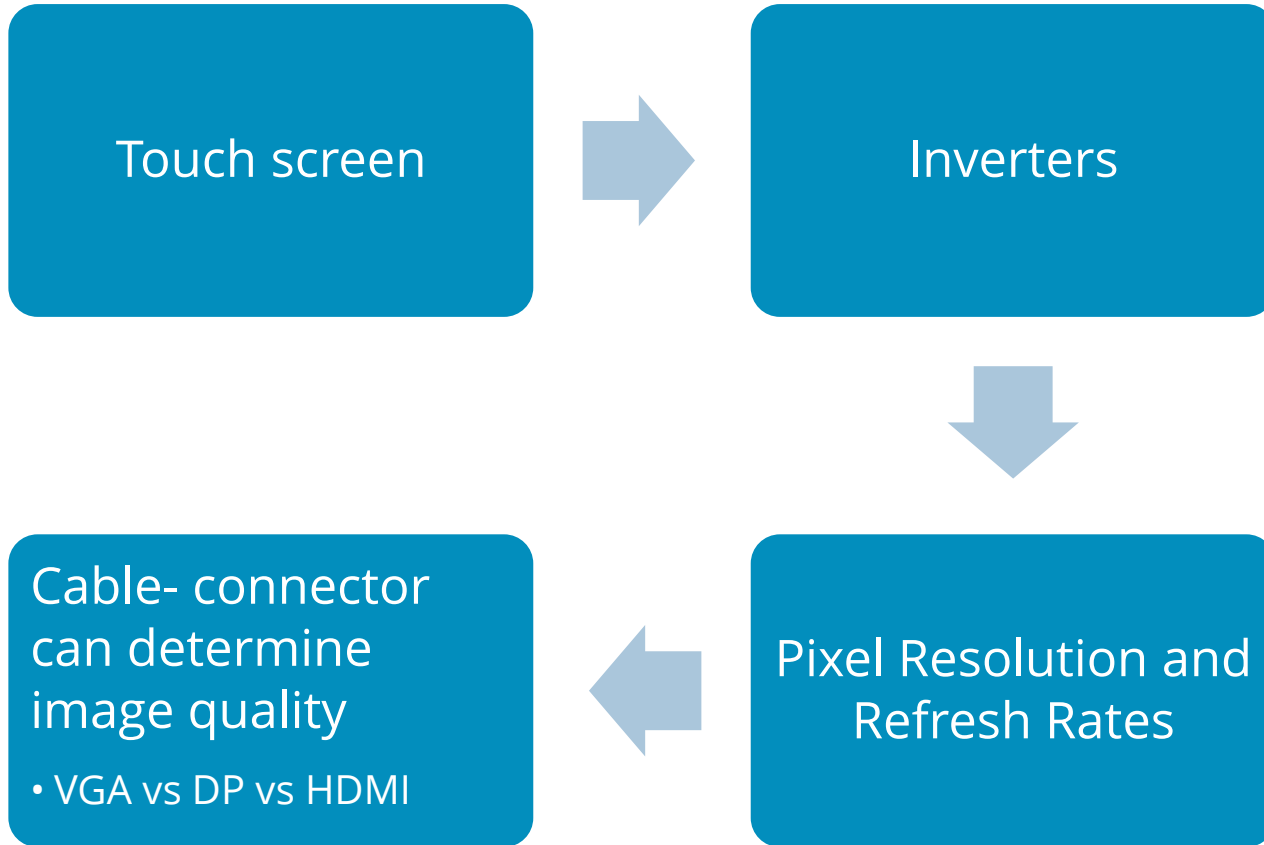
LCD (Liquid Crystal Diode) Monitors
Require a Backlight (Inverter)
Image produced by IPS, TN or VA

LCD- IPS, TN, VA

OLED
(Organic Light Emitting Diode Monitors)
No Backlight Needed

OLED vs Mini-LED

Display Components

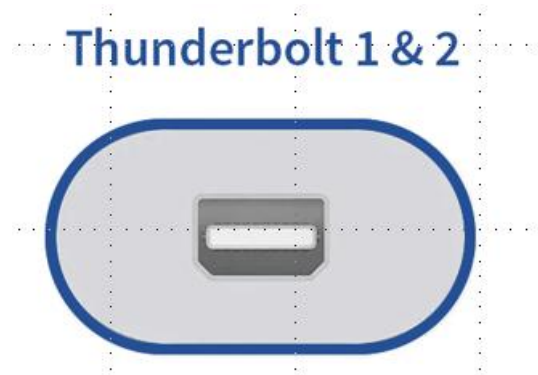


HDMI and DisplayPort Video Cables

- HDMI
 - Audio and Video Transfer
 - Versioning- High quality resolutions, color contrast, and refresh rates
 - Single Cable vs Multi cable connection
- DisplayPort
 - Audio and Video capable
 - Professional quality resolutions and high refresh rates
 - Popular with Gamers and Graphic Designers
 - Multiple Displays with single connector- Daisy Chaining
 - Sync Video and Audio

Thunderbolt Interface

- Version 1 and 2
 - Used Mini-DP interface
 - Up to 20 Gbps and daisy chaining



Thunderbolt Interface

- Version 3- USB-C interface
 - 40 Gbps over .5m cable
- Version 4- USB-C interface
 - 40 Gbps
- Version 5- USB-C Interface
 - High Speed connections for future use

Thunderbolt 3 & 4

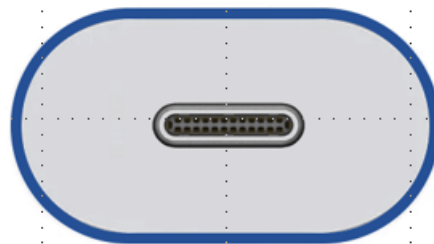


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Lightning Interface



Image ©123RF.com

- Apple proprietary connector
- iPhone and iPad
- Being replaced by USB-C connectors



Standard for Internal Storage



Smaller connector and
faster data transfer

Revision 1- 1.5 Gbps (150MBps)

Revision 2- 3 Gbps (300MBps)

Revision 3- 6Gbps (600MBps)

Serial Advanced Technology Attachment Interface (Slide 2 of 3)

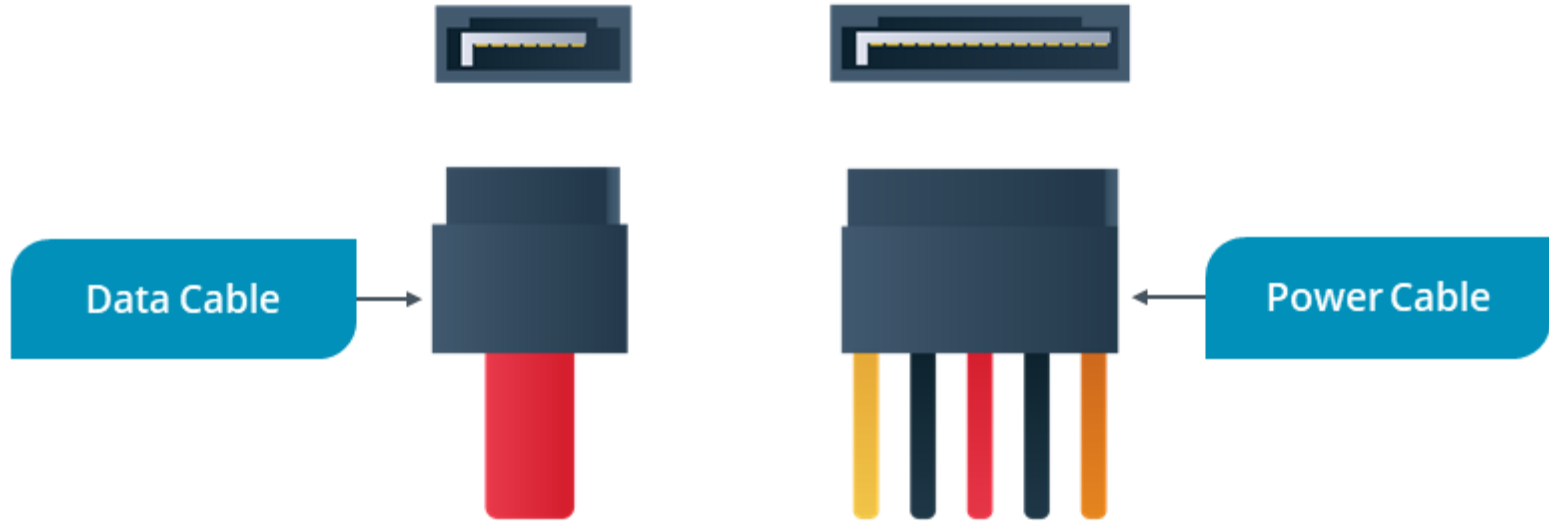


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Serial Advanced Technology Attachment Interface (Slide 3 of 3)

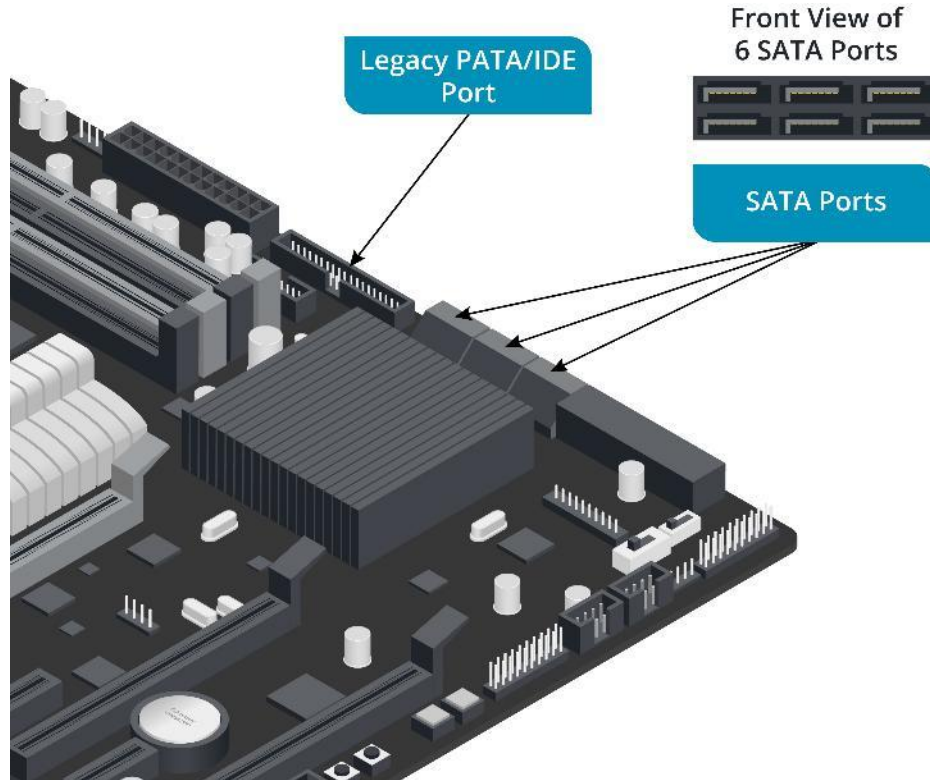


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Molex Power Connectors

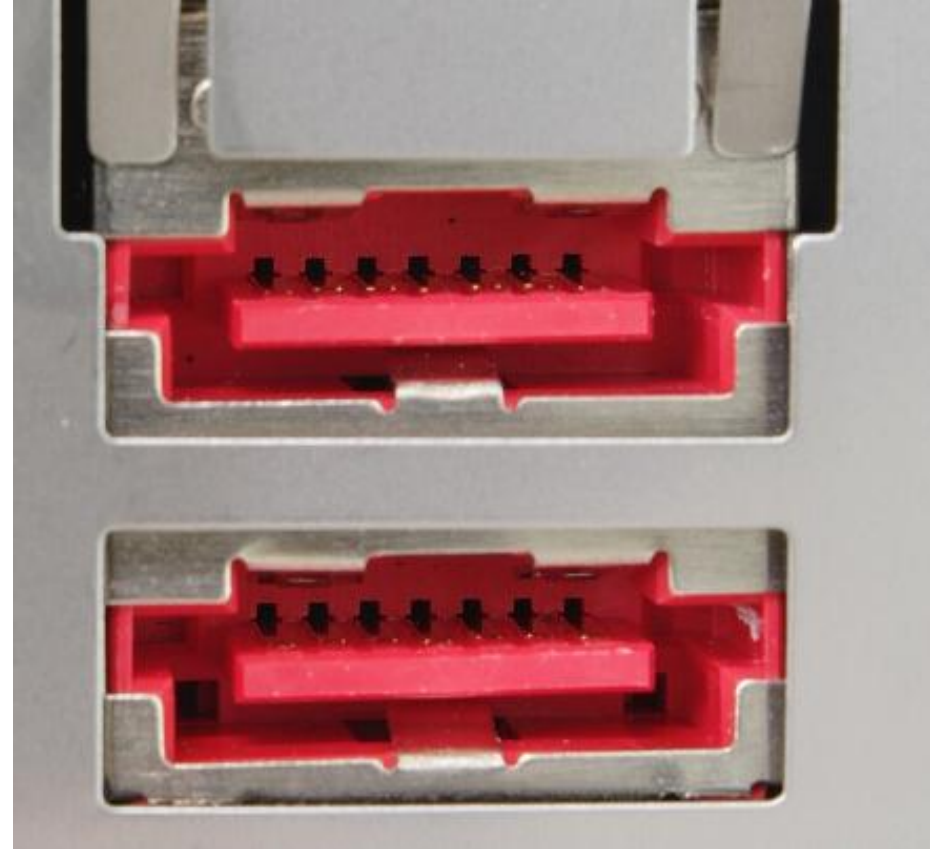


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- 4 pin connector
 - Red- 5 VDC
 - Yellow- 12 VDC
 - Black- Ground

External SATA

- Uses eSATA connector
 - Internal and External SATA cables not compatible
- Max cable length up to 2 meters



Lab Activity

- Lab: Explore the Lab Interface
(available in CertMaster Learn and CertMaster Perform)
- Lab: Install USB Devices
(available in CertMaster Learn and CertMaster Perform)
- Lab: Set Up a Desktop Computer
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Cables and Connectors

- Personal Computers
- Peripheral Devices
- Universal Serial Bus Cables
- USB Standards
- Display Types
- Display Components
- Serial Advanced Technology Attachment Interface
- Molex Power Connectors
- External SATA
- HDMI and DisplayPort Video Cables
- Thunderbolt Interface
- Lightning Interface

Lesson 2.2

Motherboards

Motherboard Functions (Slide 1 of 2)

- Provides data and power bus connectivity between hardware components
- System Clock provides timing
- Leading Manufacturers
 - Acer
 - ASRock
 - ASUSTek
 - Biostar
 - Intel
 - MSI

Motherboard Functions (Slide 2 of 2)

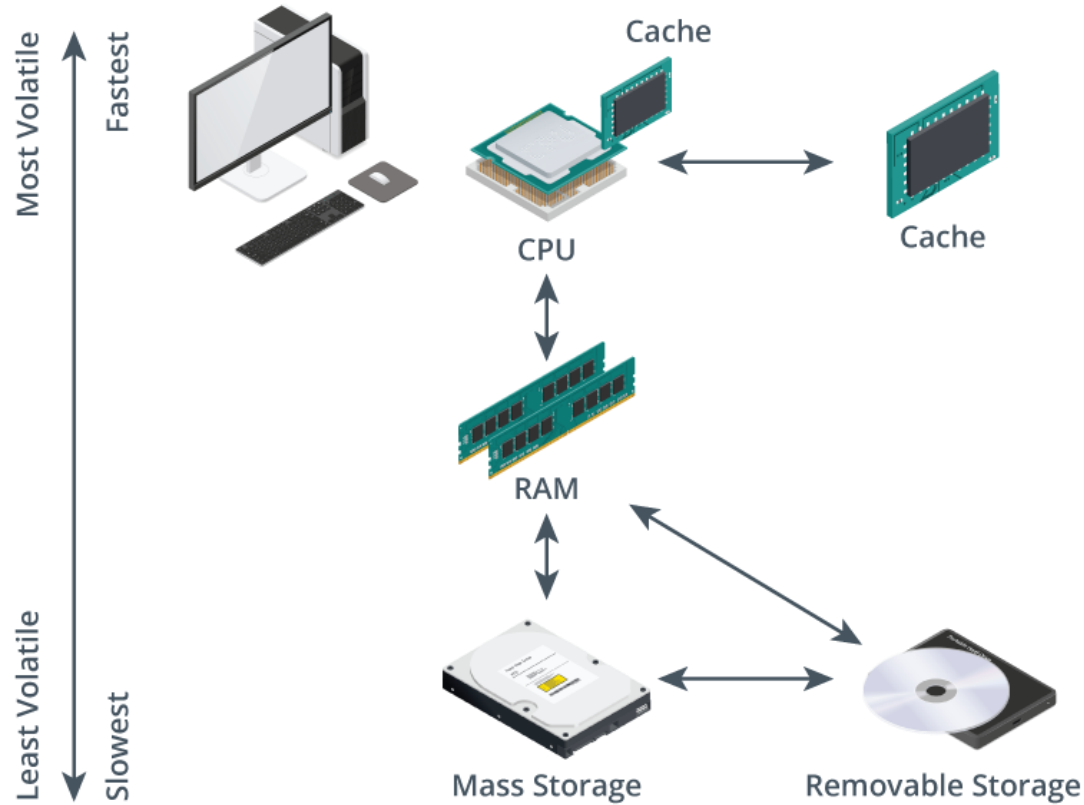


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Electrical Safety and ESD

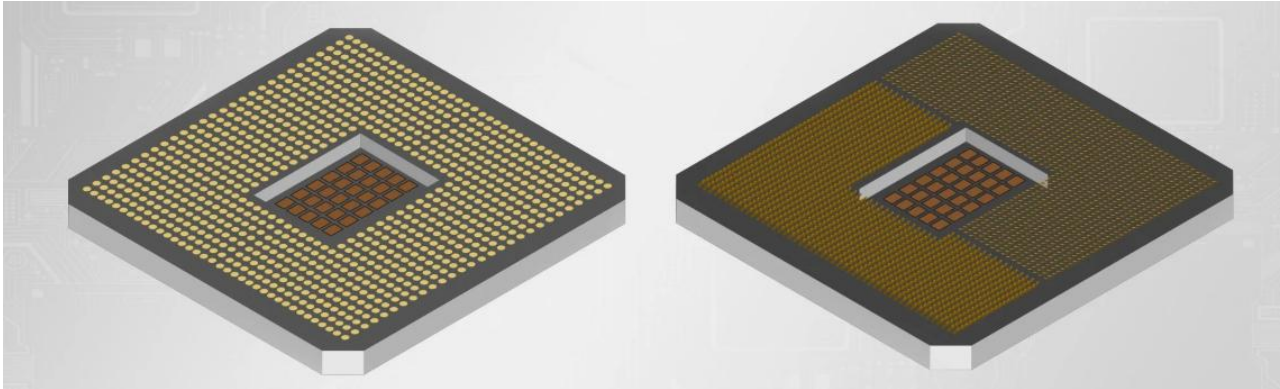
- Electrical Safety
 - Electrostatic Discharge
 - Grounding Strap and Mats
 - Electrically Safe Workbench
 - Parts stored in ESD Bags



Motherboard CPU and System Memory Connectors



CPU Socket



PGA (pin grid array)

LGA (land grid array)

Motherboard CPU and System Memory Connectors

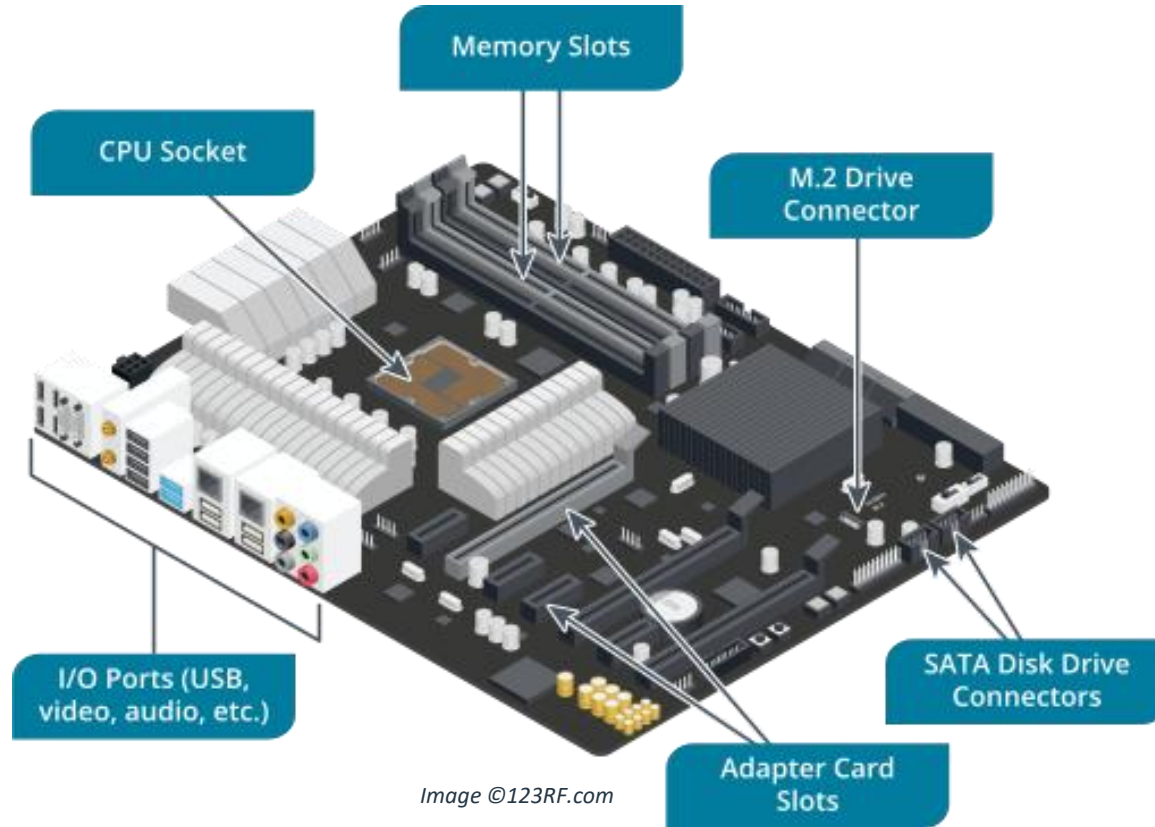


System
Memory Slots

DIMM
DDR2,
DDR3,
DDR4,
DDR5



Motherboard CPU and System Memory Connectors



Motherboard Storage Connectors (Slide 1 of 2)



A diagram of a SATA connector. It consists of a dark blue rounded rectangle with the text 'SATA' in white. To its right is a white rectangular outline representing the connector's physical shape. A thin blue line connects the left side of the blue box to the left side of the white outline.

SATA



A diagram of an M.2 connector. It consists of a dark blue rounded rectangle with the text 'M.2' in white. To its right is a white rectangular outline representing the connector's physical shape. A thin blue line connects the left side of the blue box to the left side of the white outline.

M.2



A diagram of an eSATA connector. It consists of a dark blue rounded rectangle with the text 'eSATA' in white. To its right is a white rectangular outline representing the connector's physical shape. A thin blue line connects the left side of the blue box to the left side of the white outline.

eSATA

Motherboard Storage Connectors (Slide 2 of 2)

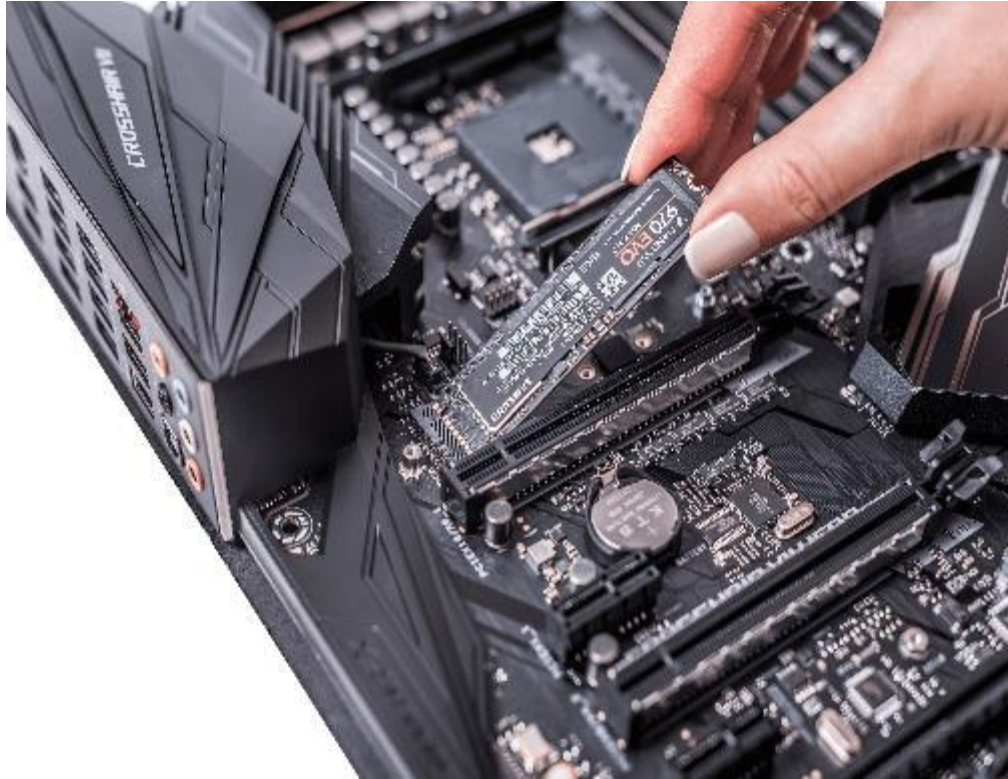
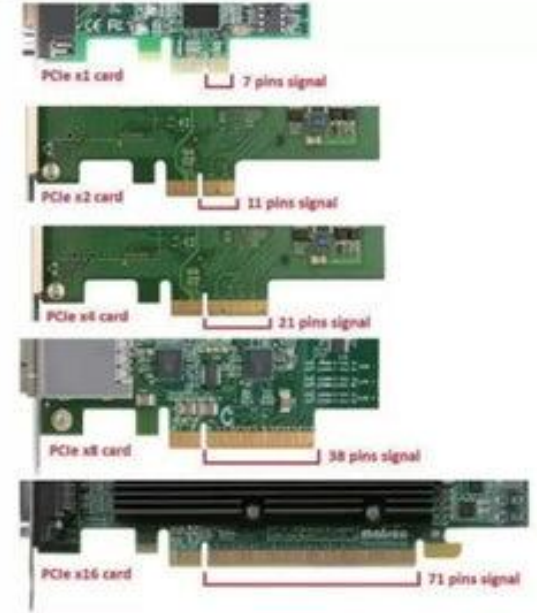
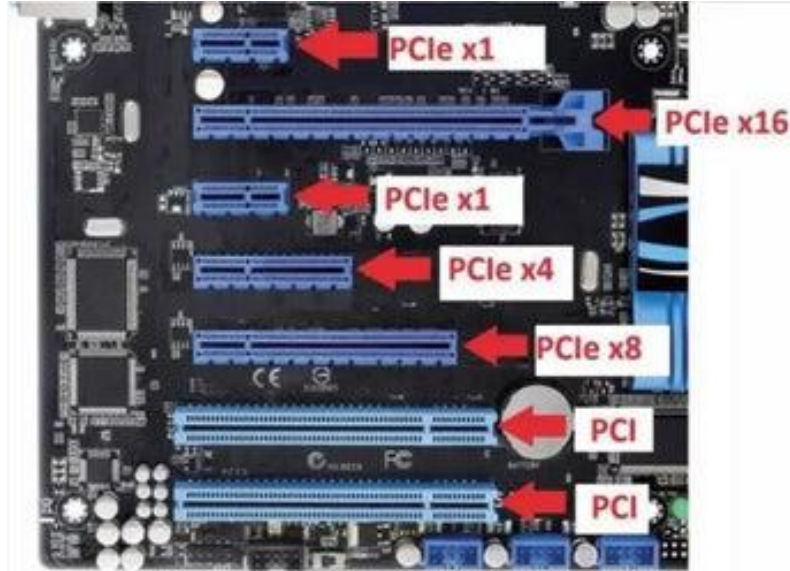


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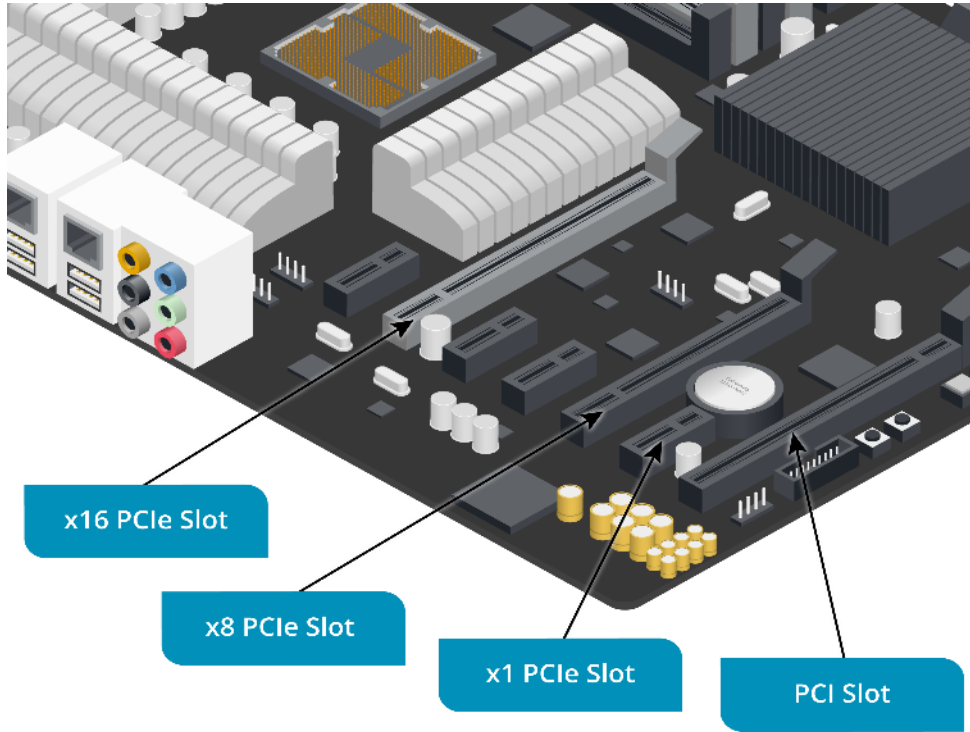
Peripheral Component Interconnect Express Interface (Slide 1 of 2)

- PCI/PCIe
 - Version 2-6
 - x1, x4, x8, x16
- Backwards compatibility



PCIe x1 card can be inserted and will work correctly in a x1, x4, x8, x16 slot.
PCIe x2 card can be inserted and will work correctly in a x4, x8, x16 slot.
PCIe x4 card can be inserted and will work correctly in a x4, x8, x16 slot.
PCIe x8 card can be inserted and will work correctly in a x8, x16 slot.
PCIe x16 card can be inserted and will work correctly only in a x16 slot.

Peripheral Component Interconnect Express Interface



PCI - Legacy Connector
Not Compatible with PCIe slots

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Peripheral Component Interconnect Interface

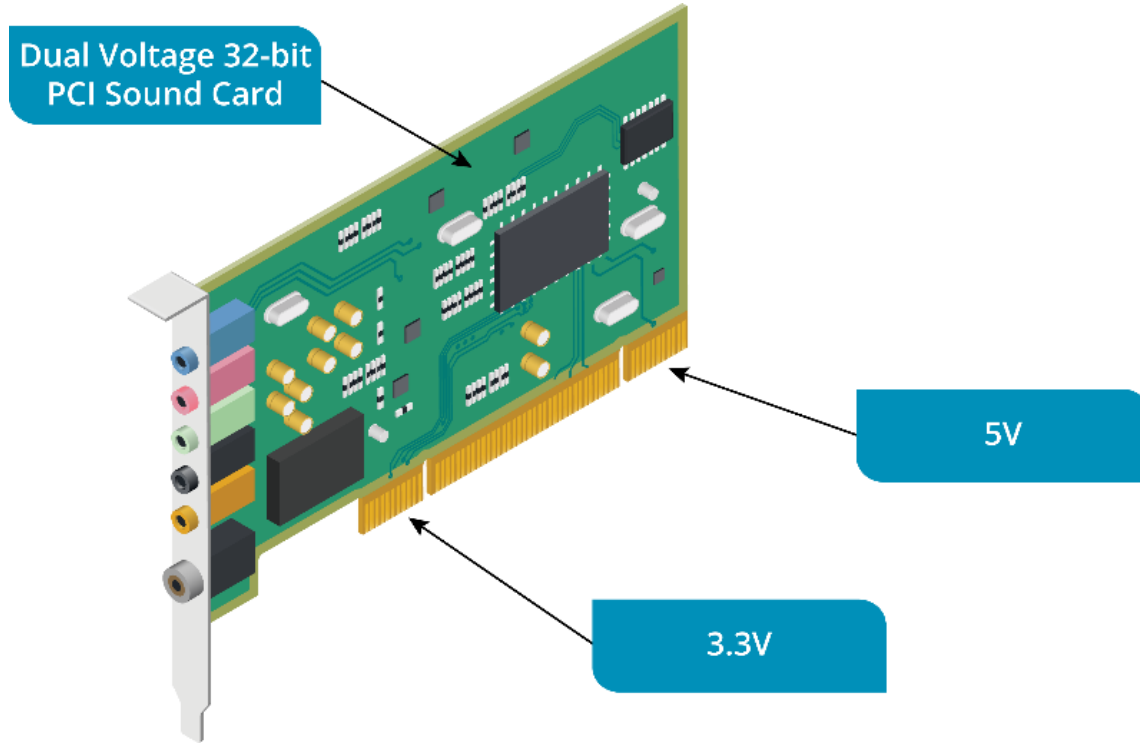


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Motherboard Form Factors

- ATX
 - 12" x 9.6" (305 x 244 mm)
- mATX (micro)
 - 9.6" x 9.6" (244 x 244 mm)
- Mini-ITX
 - 6.7" x 6.7" (170 x 170 mm)

Motherboard Installation (Slide 1 of 2)

1. Review documentation
2. Install the I/O shield
3. Insert Standoffs
4. Pre-Install the CPU and Memory
5. Align and Secure the Motherboard
6. Final Assembly
7. Cable Management

Motherboard Installation (Slide 2 of 2)



Image courtesy of CompTIA

Motherboard Headers and Power Connectors (Slide 1 of 2)

- Headers
 - Power
 - HDD activity lights
 - Audio ports
 - USB ports
- Power Connectors
 - Main P1
 - Fan Connectors

Motherboard Headers and Power Connectors (Slide 2 of 2)

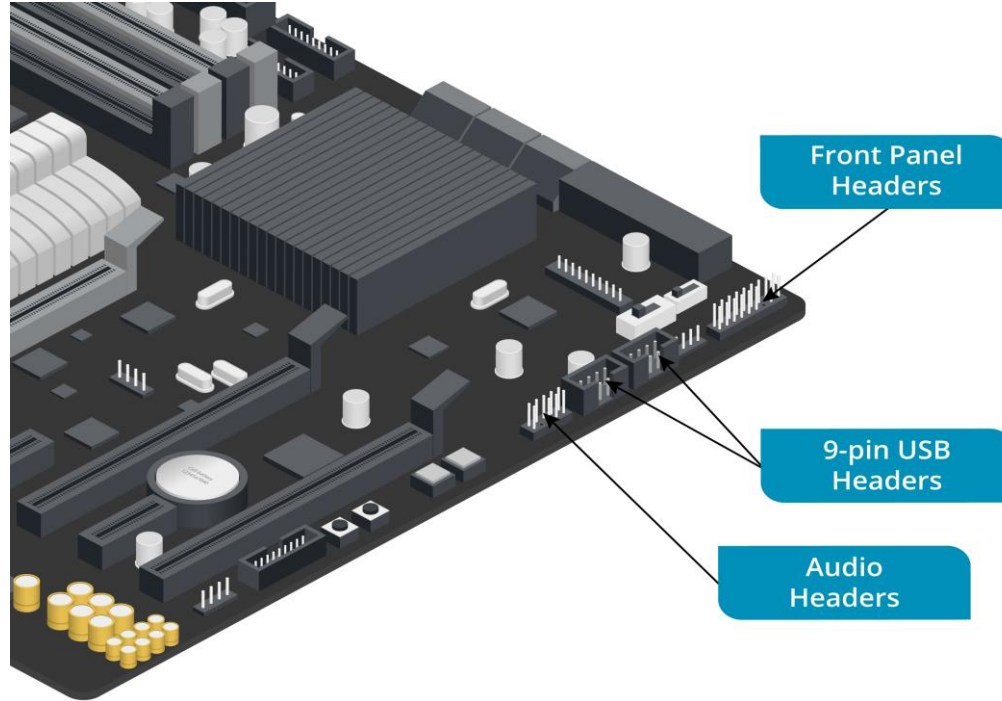


Image courtesy of CompTIA

Video Cards (Slide 1 of 2)

- Expansion Cards
- GPU
- Dedicated Graphics Memory
- Video Ports

Video Cards (Slide 2 of 2)

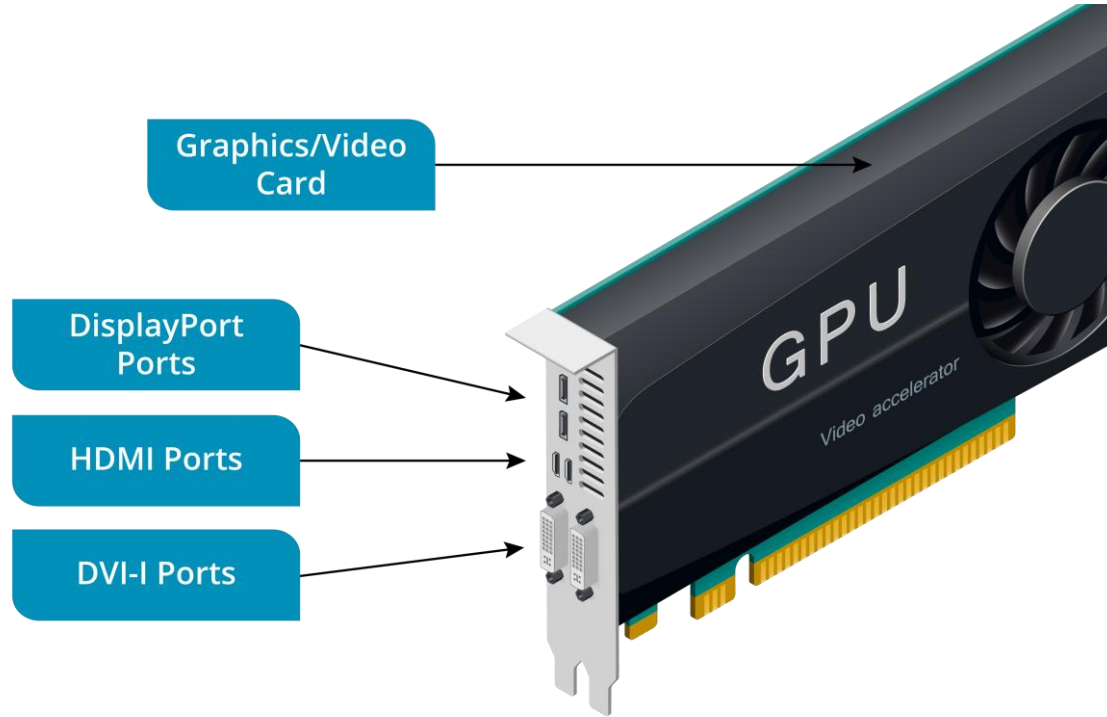


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Capture Cards

- Capture video feeds
 - Gaming
 - Designed to record or stream gameplay footage.
 - Can capture video from PCs or game consoles via HDMI.
 - HDMI streaming
 - Record video from various HDMI sources, such as game consoles, camcorders, and security cameras.
 - Used for live streaming, video production, and content creation.
 - TV Tuner Cards
 - Receive and record video from broadcast TV sources.
 - Allow users to watch and record live TV on their computer.
- Internal vs External

Sound Cards

Audio jacks- 3.5mm

Green- Audio Out

Black- Audio Out

Blue- Audio In

Pink- Audio In (Microphone)

Orange- Audio Out (Subwoofer)

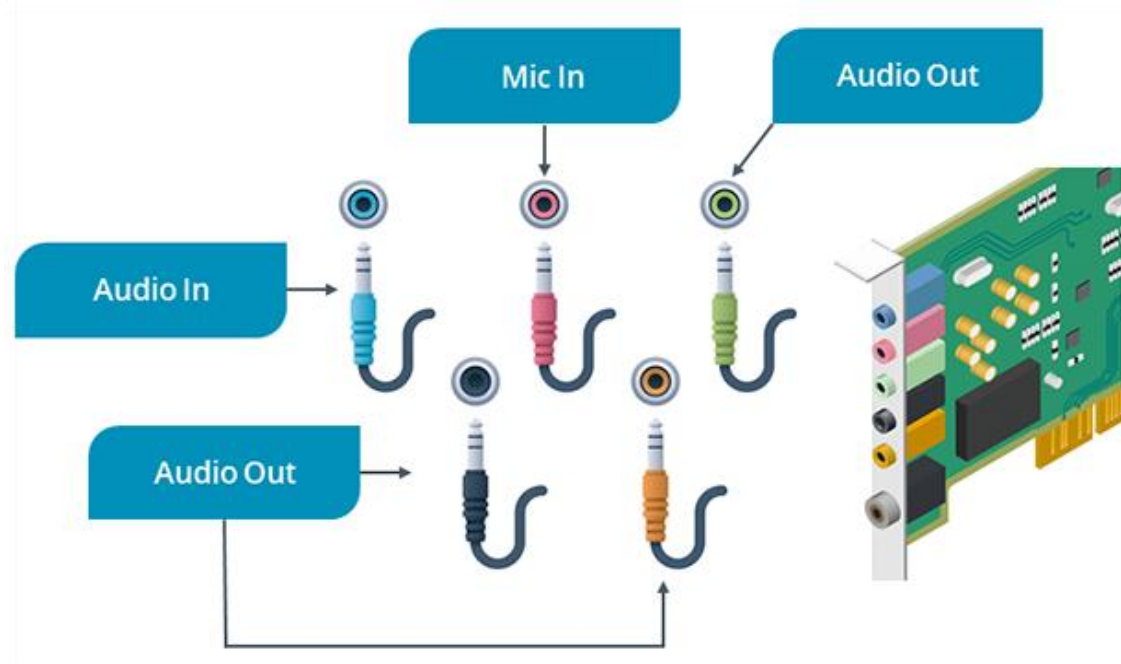


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Network Interface Cards (Slide 1 of 2)

- Copper
 - RJ-11
 - RJ-45
 - Coaxial
- Fiber Optic
- Wireless

Network Interface Cards (Slide 2 of 2)

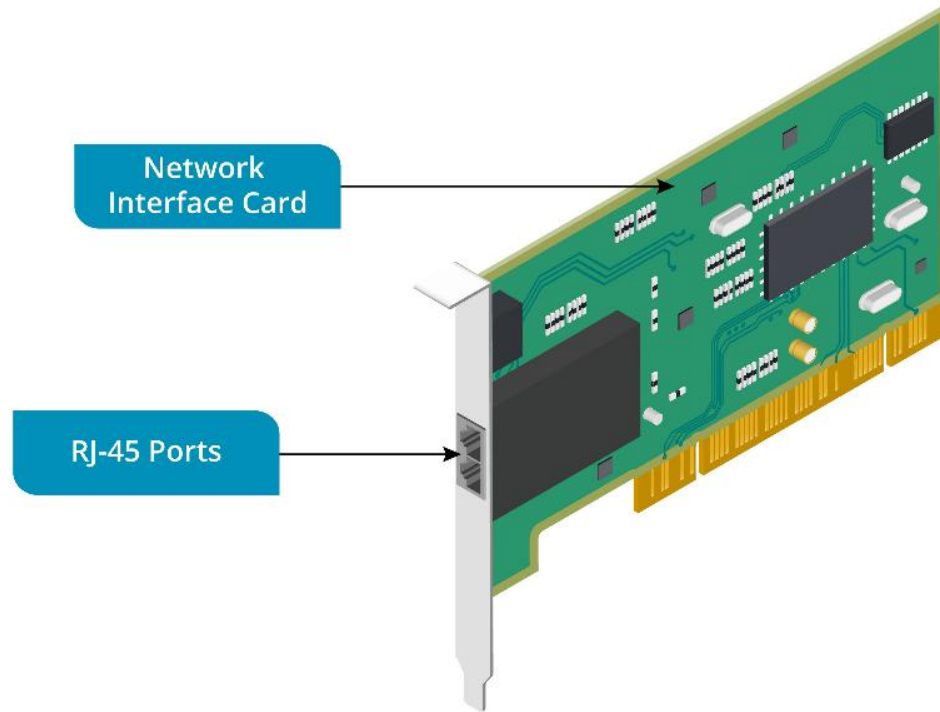


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Lab Activity

- Lab: Choose and Install a Motherboard
(available in CertMaster Learn and CertMaster Perform)
- Lab: Upgrade a Video Card
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Motherboards (Slide 1 of 2)

- Motherboard Functions
- Electrical Safety and ESD
- Motherboard CPU and System Memory Connectors
- Motherboard Storage Connectors
- Peripheral Component Interconnect Express Interface
- Peripheral Component Interconnect Interface
- Motherboard Form Factors
- Motherboard Installation
- Motherboard Headers and Power Connectors

Lesson Review: Motherboards (Slide 2 of 2)

- Video Cards
- Capture Cards
- Sound Cards
- Network Interface Cards

Lesson 2.3

Legacy Cables

DVI and VGA Video Cables (Slide 1 of 3)

- Digital Video Interface
- Video Graphics Array- Analog Serial Video

DVI and VGA Video Cables (Slide 2 of 3)

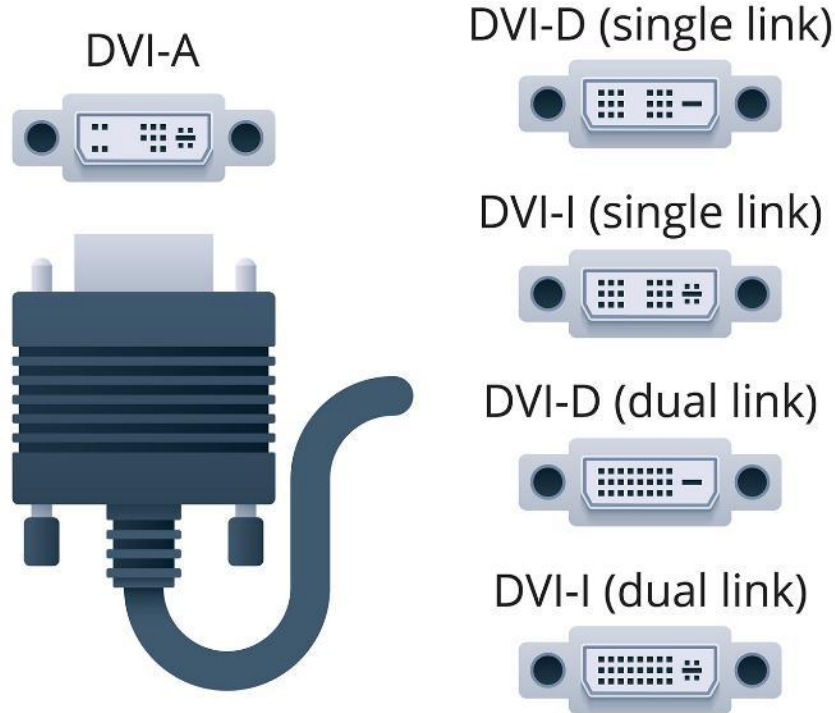


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DVI and VGA Video Cables (Slide 3 of 3)

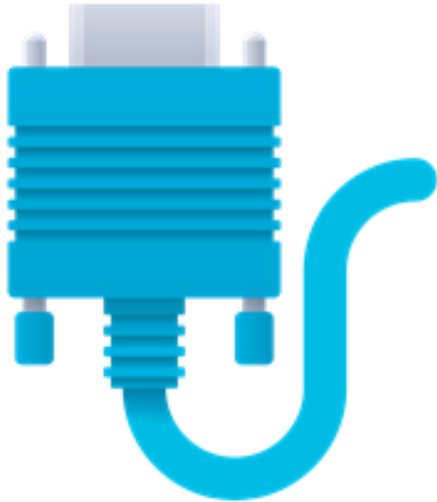


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Serial Cables

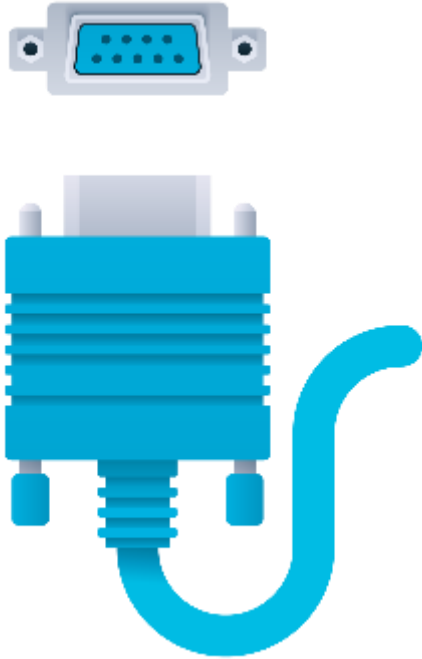


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- RS-232 standard
- DB-9 used vice the DB-25
- Windows COM Port

Adapter Cables

- Active
- Passive
- Video adapters
- USB adapters
- USB hubs
- USB adapters to other interfaces

Lab Activity

- Lab: Select and Configure Dual Monitors
(available in CertMaster Learn and CertMaster Perform)
- Lab: Set up an A/V Workstation
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Legacy Cables

- DVI and VGA Video Cables
- Serial Cables
- Adapter Cables